Tube Channel User Guide

Quick reference for the channel strip, its controls, operating modes, and signal path.

Release 1.2.3 | VST3, Audio Unit, CLAP, LV2, AAX, and standalone formats where supported.



Tube Channel combines an input and preamp, compressor, four-band equalizer, and output section. The digital and analogue-style layouts share the same controls and audio processing. Use the System page for quality, oversampling, stereo, level alignment, and circuit settings.

Main page



Digital main page, showing the input, compressor, equalizer, and output sections



Analogue-style main page, showing the same controls in a rack layout

Toolbar

Control	What it does
Init	Opens the factory-preset menu. Init restores the starting settings; the other entries are starting points for vocals, bass, drums, tube drive, and limiting.
A / B	Switches between two parameter snapshots.
Copy	Copies the current snapshot to the other snapshot.
Undo / Redo	Reverses or reapplies the last control changes.
CH A / CH B	Selects the channel controls when Dual Mono is enabled. These tabs are hidden otherwise.
System	Opens the System page. Press it again to return to the main page.
Analogue / Digital	Changes the interface layout. It does not change the audio processing.

Input and preamp

Control	What it does
Mic / Inst	Chooses the microphone or instrument input path.
Gain	Sets input gain. The range is approximately 0 to 40.6 dB.
+20 dB	Adds an additional 20 dB of input gain.
+48 V	Turns phantom power on or off. Use it only when the connected microphone requires it.
Phase	Reverses signal polarity.
Impedance	Sets the input load from 150 ohms to approximately 3.4 kilohms.
Low Cut	Sets the low-frequency cut from approximately 11 Hz to 250 Hz.
Preamp / Preamp Out	Sets the level leaving the preamp; the control can mute the preamp output.
Plate Hi / Norm	Selects the higher or normal internal tube supply setting.
Input Trim *(System page)*	Adds or removes up to 24 dB before the channel processing.

Compressor

Control	What it does
Active / Bypass	Enables or bypasses the compressor.
Threshold	Sets the compressor threshold, displayed in dBm.
Ratio	Sets the labelled ratio from 2:1 to 7:1. Click the Ratio control to switch the brickwall limiter on or off.
Attack	Sets how quickly compression responds, from about 4 ms to 33 ms.
Release	Sets how quickly gain reduction recovers, from about 0.1 s to 1.1 s.
Auto / Manual	Selects automatic or manual detector timing.
Int SC / Ext SC	Selects the internal detector signal or the host's external sidechain input. The host must provide a sidechain bus for Ext SC.
Output	Sets compressor make-up/output level.
Gain Reduction meter	Shows the compressor's current gain reduction.

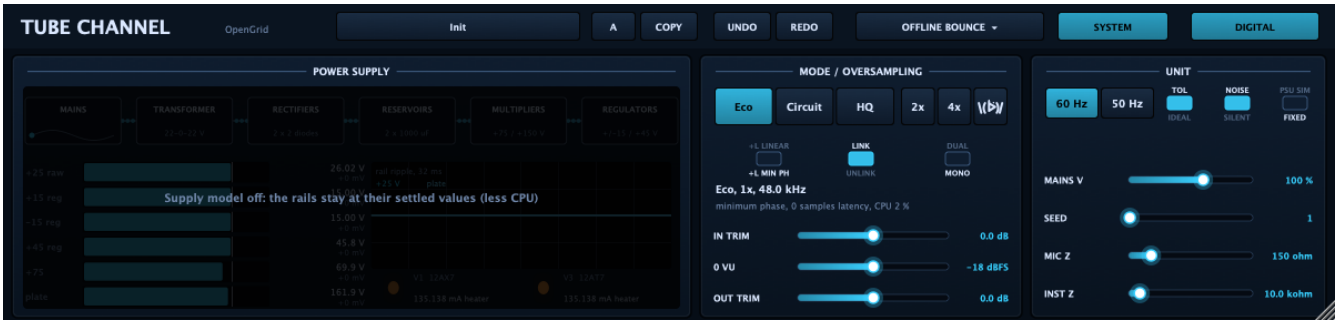
Equalizer

Control	What it does
EQ In / EQ Out	Enables or bypasses the equalizer.
Low	Boosts or cuts the low shelf, up to 12 dB in either direction. **40 Hz / 120 Hz** selects its frequency.
Low-Mid	Sets the low-mid band's boost or cut. **LM Freq** sets its centre frequency; **LM X10** selects the higher frequency range; **Narrow / Wide** changes its bandwidth.
High-Mid	Sets the high-mid band's boost or cut. **HM Freq**, **HM X10**, and **Narrow / Wide** set its centre frequency, range, and bandwidth.
High	Boosts or cuts the high shelf, up to 12 dB in either direction. **6 kHz / 18 kHz** selects its frequency.

Output and meters

Control	What it does
Master	Sets the final output level.
Output Trim *(System page)*	Adds or removes up to 24 dB after the channel processing.
+4 dBu / -10 dBV	Selects the nominal output and meter reference level.
PRE / COMP / EQ / OUT	Selects the point in the signal path shown by the VU meter.
VU meter	Displays the selected signal level. The clip lamp indicates output clipping.
Preamp Tube Warmth meter	Shows the tube-stage drive activity.

System page



Digital System page, with the quality, oversampling, unit, and supply settings



Analogue-style System page with the same settings

Modes, oversampling and stereo

The engine buttons are ordered **Eco | Circuit | HQ**.

Control	What it does
Eco	New-instance default. Uses calibrated filters and nonlinear maps, residual ADAA and no circuit solves. Heavy-drive behavior can differ from Circuit/HQ.
Circuit	Renamed Default mode. Retains the circuit audio stages and prototype compressor detector.
HQ	Retains the circuit audio stages and original modeled compressor detector.
2x / 4x / Derp symbol	Independent oversampling at exactly 2x, 4x or 8x, at any session rate and with any engine. Click the lit button again to switch off (1x, no button lit). New instances start at 1x.
+L Min Ph / +L Linear	Chooses minimum-phase IIR or linear-phase FIR oversampling filters. Linear phase reports latency when oversampling is active.
Link / Unlink	Links or separates the left and right channel controls.

Control	What it does
Mono / Dual	Selects a single linked stereo path or independent Dual Mono channel settings. In Dual Mono, use CH A and CH B to edit each side.

The status text reports the engine, effective factor/internal sample rate, phase, latency and CPU use. Oversampling increases CPU cost; Eco's published speed results use 1x. Eco uses reduced supply/load states and seeded gain/noise variation rather than per-component circuit-equivalent behavior.

The existing three-choice Quality automation range is preserved. Eco and independent Oversampling have separate automation parameters. Loading a pre-Eco state keeps Eco off. Loading a state from before the independent oversampling control preserves its previous adaptive factor; the first engine click freezes that factor, so subsequent mode changes keep oversampling unchanged.

Offline Bounce

The **Offline Bounce** dropdown sits between **Redo** and **System** and is visible on both pages and in both skins. It sets the engine and oversampling together:

Engine	2x	4x	Derp (8x)
Eco	Eco | 2x	Eco | 4x	Eco | Derp (8x)
Circuit	Circuit | 2x	Circuit | 4x	Circuit | Derp (8x)
HQ	HQ | 2x	HQ | 4x	HQ | Derp (8x)

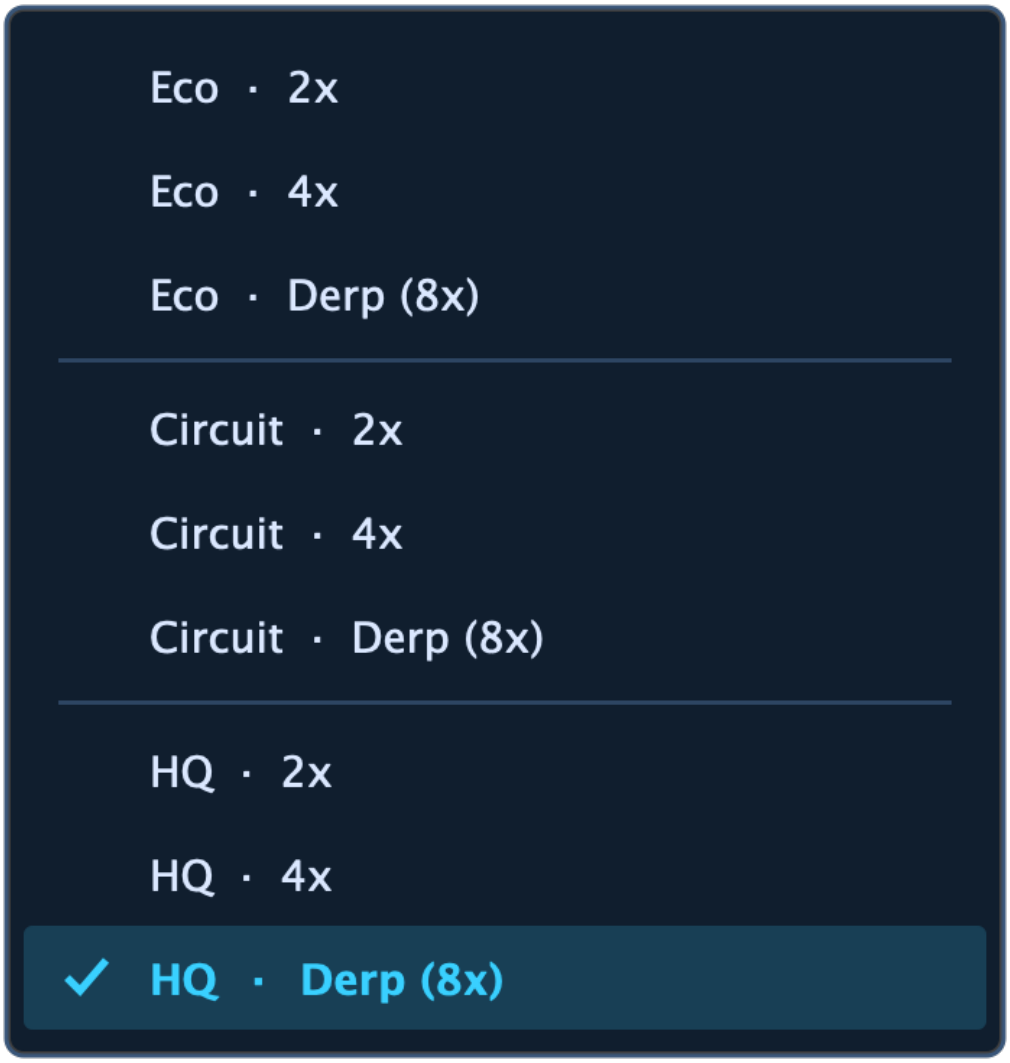
New instances default to **HQ | Derp (8x)** for offline bounce, with **Eco / 1x** for normal work. The override applies only when the host explicitly reports offline rendering. **Realtime exports are realtime**: they use the normal playback settings, just like playback and recording. There is no manual force-offline switch.

The dropdown does not alter playback mode/oversampling automation. Both settings are saved in the project. Sessions saved before 1.2.1 continue to use their playback settings for offline bounce until a dropdown option is chosen. Its tooltip explains this compatibility state. The nine choices are exposed to the host as the appended Offline Bounce parameter; the accompanying override flag preserves older sessions.

The processing path changes before the first nonempty render block, without the live mode-change fade. The working engine is retained separately, and repeated bounces restart the offline engine consistently. Hosts that announce rendering or restore parameters at the first offline block are supported without relying on the UI message loop. Such offline-only preparation may increase export startup time; playback callbacks do not build or destroy engines.

Minimum-phase mode reports zero oversampling latency. With **+L Linear** and the offline override enabled, a fixed latency budget covers the available factors. The shorter path is delayed to match it, keeping host timing unchanged when an offline render starts or ends. This can add latency to Eco/1x while Linear is on. The global phase setting, gain trims, sidechain and stereo/Dual Mono settings apply to both paths. Changing the bounce combination during playback preserves the working engine's filter/detector history. The dropdown lights while its offline path is active.

The Eco/Circuit/HQ buttons continue to show the saved **playback** selection during a bounce. The status footer reports the active offline model, but fast renders or hosts using a separate render instance may not redraw the editor.



Nine Offline Bounce combinations, menu illustration

Menu illustration. Native popup spacing and highlighting follow the platform skin.

Unit and power supply

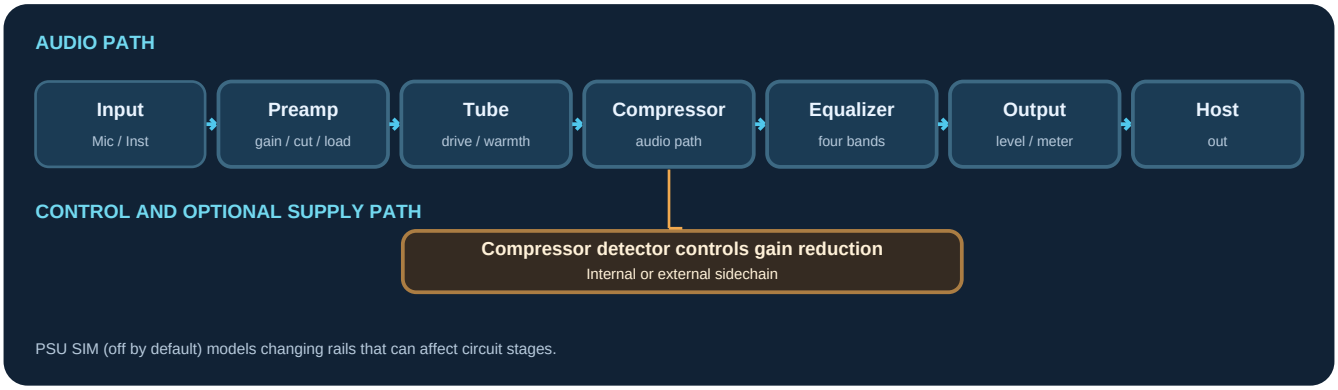
Control	What it does
60 Hz / 50 Hz	Selects the modeled mains frequency.
TOL / IDEAL	Enables per-unit component tolerances or uses ideal nominal values.
NOISE / SILENT	Enables or disables circuit noise.
PSU SIM / FIXED	Enables the power-supply circuit simulation or holds its rails at settled values. New instances start with PSU SIM off to save CPU. A saved session restores its saved setting.
Mains V	Sets the modeled mains voltage from 85% to 110%.
Seed	Selects a repeatable component-tolerance/noise seed.
Mic Z	Sets the modeled microphone source impedance from 50 to 600 ohms.
Inst Z	Sets the modeled instrument source impedance from 1 to 100 kilohms.
Power Supply display	Shows the modeled supply rails and their ripple when PSU SIM is enabled.

Level alignment

Control	What it does
In Trim	Sets input trim from -24 to +24 dB.
0 VU	Chooses the digital level that corresponds to 0 VU: -14, -16, -18, -20, or -22 dBFS.
Out Trim	Sets output trim from -24 to +24 dB.

Signal flow

The main audio moves through the channel in this order. The detector listens to the selected internal or external sidechain and controls compressor gain reduction. The supply simulation, when enabled, calculates changing supply rails that can affect the circuit stages.



Tube Channel audio and supply signal-flow diagram

Notes

- The plugin's first/default state has PSU SIM off. Turning it on uses more CPU.
- In a saved project, the host-restored parameter value takes precedence over the new-instance default.
- If you hear crackling or breakup, first check the host's audio buffer and real-time CPU load, then try Eco mode and turn off PSU SIM.
- Source code is MIT licensed. All official plugin binaries, including AAX in the macOS installer, are freeware. The installer includes the applicable third-party license texts. See LICENSE and THIRD_PARTY_LICENSES.md.